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Speculative frictions of cohabitation: Storytelling and diegetic prototyping in more-than-human architecture

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ABSTRACT: Narratives of "domestication" determine which creatures are accepted or not within human spaces, nonetheless, more-than-human anthropology indicates that, in different contexts, that distinction can sometimes blur, suggesting spaces supporting alternative human-nonhuman relationships. Experiments of multispecies architecture often don't prioritise interactions with humans, but it's precisely in the spaces where different species co-exist that design has opportunities to look for alternative hierarchies, gaining a broader agency: conflicts and frictions can become particular moments of inquiry. This paper contextualizes and critically discusses concepts related to three speculative projects by Animali Domestici, which investigate coexistence in mundane everyday spaces through techniques of storytelling and diegetic prototyping, whose goal is to "situate" within the speculative scenario, aiming at focusing on nuances of friction. These projects engage both with physical and digital prototypes, but they also explore their hybridization and contamination as an opportunity to blur the boundary between reality and fiction.

1 INTRODUCTION

This paper is aimed at contextualizing and reflecting on some experimental processes used by the authors' practice, Animali Domestici, to explore the field of multispecies coexistence in architectural design in three specific projects. The contextualization is both thematic, looking at domestication and coexistence, and methodological, with notes on diegetic prototyping and anticipatory ethnography.

Imagining alternative forms of coexistence requires first framing the different notions of multispecies relationships from the perspective of spatial design, as addressed in section 2. This requires undoing conventional notions of domestication, taking inspiration from existing alternative arrangements, and noticing some of their characteristics.

Section 3 gives a glimpse of the multispecies spaces that are currently investigated by design and architecture, understanding the challenges they present. Here there is a tendency to overtly simplify nonhuman relationships and a risk of reducing them to behavioural patterns and functional issues, generally overlooking subtle nuances, crucial in fragile equilibriums.

In the following section, a focus on frictions helps concentrating on mutualism and interactions, particularly relevant in spaces of everyday coexistence, which are rarely considered, as difficult to implement.

Speculative practices can be useful in this sense if they can achieve a situated approach, which could be pursued through detailed constructed spaces acting as diegetic prototypes, as discussed in sections 5 and 6. Section 7, instead, introduces the perspective of anticipatory ethnography posing the accent on how storytelling can allow to "situate via proxy".

The three projects briefly mentioned, in between the methodological sections, give a glimpse of the Animali Domestici's approach to the field, exemplifying in short both the topics and methodologies introduced, especially touching on the re-negotiation of boundaries in *KK Post-vernacular*, the situated recontextualization of design experiments in *Hardware Stories*, and both regulatory questions and the active role of storytelling in *Beyond Byggestop (Beyond Stop Building)*.

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2 DECENTERING DOMESTICATION

"Popular domestication narratives" depict a set of relationships between humans, animals, and plants that are "compelling and easy to grasp", as they "explain and order through binary coupling: the civilized from the savage, the domestic from the wild." As such, they neglect different forms of "other-than-human engagement" (Swanson et al., 2018).

Architecture plays a crucial role in such narratives, as, within them, it delimits the spatial spheres of those categories and embodies the boundaries in its very materiality. Studies on traditional "architectures of domestication" focus mainly on "complex models of mutualism and co-domestication," where "animals move between physically altered 'domestic' forms and 'wild' or feral forms" (Anderson et al., 2017). These more nuanced interpretations of the complex multi-species relationships suggest many conditions in-between categories like domestic and wild, hence suggesting the existence of relative *in-between spaces*.

To this end, "seeking out domestication assemblages that are rendered invisible or peripheral by dominant narratives," more-than-human anthropologists provide inspiring examples of alternative arrangements: in these accounts, the concept of domestication evolves towards "a multiple, mutual, and partly unintentional process" (Swanson et al., 2018).

Several animals studied by Anna Tsing in the Meratus rain forests in Borneo, for example, showcase a flexible connection to human environments. In the case of wild chickens, for instance, the opportunities to find shelter for nesting are what keeps them around houses, rather than any form of coercive enclosure.

In her essay titled "Nine provocations for the study of domestication," Tsing brings up the inspiring term "domestication-as-rewilding" to hint at the "productive confusion of domestic and wild" and, therefore to a "cospecies landscapes in which no species can be said to be in charge" (Tsing, 2018). In spatial terms, many of her examples in this argument, provide protection and places to safely find food and resources, rather than creating confinement, hence they are based more on "multispecies engagement", rather than domination.

One of the immediate effects of "decentering domestication" is that the supposed frontiers between humans and nonhumans get blurred on multiple levels (Swanson et al., 2018). This immediately prompts a compelling question for design inquiries in this field: what are the spaces and the connected practices that support the dissolution of such boundaries?

Multiple examples discussed by more-than-human anthropology suggest that it is a complex mixture of conditions, opportunities, mutual support, and much more that define those spaces.

3 MULTISPECIES ARCHITECTURE

To situate and contextualize the explorations proposed here within the heterogeneous field of nonhuman-related design, it seems useful to briefly map some of its features. Within the sparse specific literature identifiable, the concept of "multispecies" architectural design seems to offer a more suitable entry point to this inquiry, and it relates, in general terms, to a "process that considers human and non-human needs (e.g. animals, plants, and microbiota)" (Grobman et al., 2019). This goal might unfold in several different ways, but one of the most obvious and straightforward is "integrating elements of nature at the building scale," both plants and animals, restoring or creating habitats for them (Grobman et al., 2019).

Within the global climate crisis, the widespread decline in biodiversity is one of the few aspects that allows local actions to have a tangible impact. Habitat loss is one of the core phenomena affecting the biosphere and architecture has a direct effect on it, through the occupation of land, but also in its integration of potential habitats for nonhumans in the built environment, where its surfaces, like the envelope and inner materials play a crucial role as hosting environments.

Specifically looking at the integration of the animal sphere, research notes that often "viewpoints held by design stakeholders regarding animals are anthropocentric, utilitarian, subjective, contextual, and influenced by their comprehension of the significance of various species" (Grobman et al., 2019). How could we start flipping these perspectives and foregrounding the nonhuman side as a base for multispecies design?

Inquiring about their characteristics from a nonhuman perspective and the opportunities that hosting environments can provide nonhumans is crucial to understanding where and how multispecies spaces could be generated and what kinds of interactions they could support. As an example, "building envelopes that can host cavity-dependent animal species," like insects, have been studied in computational design research, because of the specific interest in the complex geometries.

Most literature considered hints that many of the relatively few experiments centered around the perspective of animals are primarily concerned with their habitat only, rather than with the mutual interaction with humans, as in the case of many standalone installations and pavilions, that are often unfolding in spaces where human occupation is sparse.

There don't seem to be comprehensive accounts of the field to date, but one of the most established examples in multispecies architectural design, recalled in most publications, is the work of Ants of the Prairie. Their "Bat tower" (2010) utilizes multiple strategies to create new habitats, through form and materiality, for instance, as "ribbed construction includes a series of 'landing pads' near the top of the tower [and a] pattern of grooves … to more easily climb." But, as discussed with alternative domestication, the provision of prey and resources can be strategic, as in this case "the site boasts an abundance of mosquitoes and other bat-attracting insects", but also "chives, oregano and other bat-attracting herbs planted within the base of the tower," suggesting the strategic admixture of existing and *designed* conditions (Hwang, 2010).

4 SPACES OF COEXISTENCE

Nevertheless, standalone multispecies experiments present the risk of applying yet another form of autonomy, reducing substantially the potential to undermine established hierarchies. In this sense, it's in the overlapping spaces of coexistence that design inquiry can explore alternative forms of mediation, as "cohabitation [. . .] means 'living with'—something that is not always pleasant, innocent, beautiful, or free of danger" (Amir, 2021).

One of the most valuable fields of exploration might be the areas of overlap, where humans and nonhumans interact, even though many times just indirectly, as in the examples of alternative domestication recalled earlier, facilitating opportunities and conditions for protection and nourishment. It is, therefore, necessary to broaden the spectrum of types of relationships, where negotiations don't necessarily lead to conflicts, but are also very different from an everbright harmony, often portrayed by the *entourage* of commercial design imagery.

Here "friction" becomes a useful term as it highlights the agency of "encounters across difference", which is how anthropologist Anna Tsing refers to frictions in her homonym book that utilizes the concept of interactions on a global scale (Tsing, 2005).

In physics, the term "friction" relates to the action of resistance produced between surfaces and it is inherently related to mutual interactions. Its meaning in physics is a reaction force generated by contact, but not an impact. Its comparison with words like clash or scratch, suggests its subtle degree of contraposition, clarifying its metaphorical use for small conflicts. All these elements, interaction, difference, and subtle contraposition, make it a very suitable concept to address the reactive nature of these multispecies relations of coexistence. Tsing also reminds us that the "effect" of such encounters "can be compromising or empowering", and that's also where design can find some agency in this sense (Tsing, 2005).

5 RENEGOTIATING THE BUILT ENVIRONMENT: KK POST-VERNACULAR

The project "KK Post-vernacular", developed by Animali Domestici for Seoul Biennale of Architecture and Urbanism 2021, is a set of hacks for details of ordinary houses in Thailand, that are aimed at providing hosting environments for other beings, like insects or bugs of various kinds, within modified architectural components.

Developed with Thai carpenter and timber expert Suntan Viengsima, it looks into the different relations that insects have with various types of timber, depending on the substances that they contain or their cellular structure. Teak timber, for instance, is widely used in construction, because it

contains oils that naturally repel termites, but Makha, instead, can be enjoyably appropriated by various insects.

The details were studied differentiating structural and non-structural components, but also integrating additional sacrificial layers as buffers and to create hosting cavities.

Crafting a micro-landscape on top of the built environment, the project suggests the possibility of supporting urban biodiversity and allowing creatures to inhabit the surface of architectural spaces.

The hacks were developed with the intention to be easy and intuitive to implement in a DIY form, with the tools available in most houses and the eventual help of the frequent handymen.

Some of the material manipulations incorporated multiple inspirations and influences: the traditional carpentry practice of sealing the end of timber logs with oil-based or enamel paint to slow the moisture release, for instance, was substituted with holographic paint popular in car tuning to appeal to insects attracted by iridescent surfaces.

Other elements, instead, were more directly mimicking natural environments, as the roof tiles replicated with tree bark and infilled with mosses. These require constant slow watering to avoid drying out in the warm Thai climate, so a simple system of dripping cotton ropes was incorporated above them.

The documentation of the prototype and proposed implementation in the exhibition publication (Figure 1) intentionally mixed up altogether images of different natures: physical prototypes, material tests, pictures of existing houses, photos from the material source scrapyards, but also detailed 3d models of some details, and renderings of the reconstructed immediate context. The visual similarity and chromatic consistency of various images disguises their very different nature. This was used in an attempt to blur the perception of the unreality of the proposal, encouraging to suspend disbelief about its potential implementation.

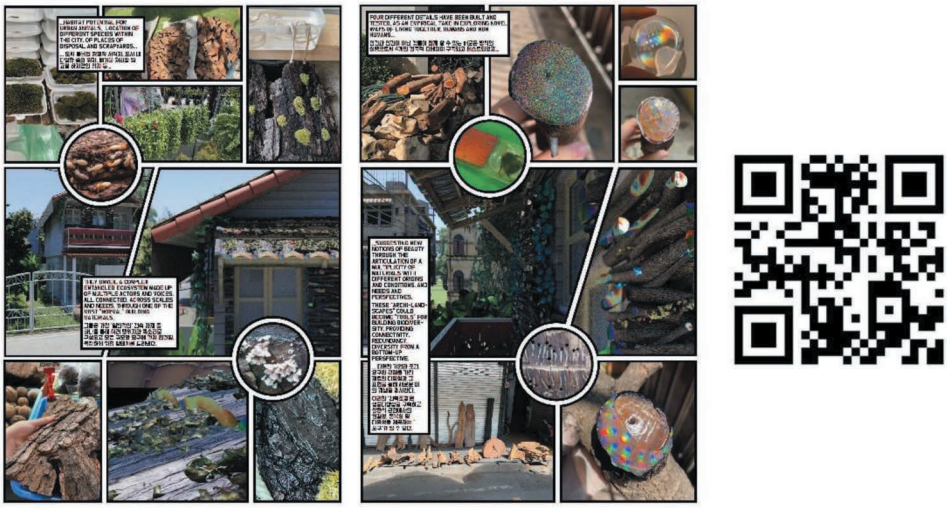


Figure 1. Pages about "KK Post-Vernacular" from the biennale publication – Animali Domestici - 2021. Source: document and images by the authors – Hi-res link QR code or: https://imgur.com/a/QOlKrHq.

6 DESIGN FICTION

The concept of speculation (and its adjacent design fiction) is often abused and misunderstood in design fields, so it can be relevant to clarify a few concepts from scratch.

If it is rather obvious to say that any object of design is unreal until it is actually materialized, but what matters in this perspective is whether the condition of "unreality" is operationalized.

This also involves embracing a more nuanced understanding of such conditions besides the binary condition real/unreal, as in the case of future projections, where the level of likelihood stands in reverse of a progressive degree of unreality of the future situation envisioned.

The so-called "cone of possibility" is a frequent diagram to visualize it, where the central portion of the spectrum represents the "probable future," which is the most likely to happen, and on the opposite side, outside the boundaries of the cone we find the realm of the "impossible future," which stands for full unreality, as a large part of Sci-fi (Dunne & Raby, 2013). The portions in between these defined conditions can be more relevant to speculative design, as their *realism* depends on multiple factors, like political or economic support, which might not be apparent.

It's the inner sector around the probable, nevertheless, that adds a crucial aspect to the equation, which is "plausibility". A plausible future scenario might not be the one that is on track to happen, but it's believable under the current conditions, hence it might be able to trigger some form of identification in the audience.

Sci-fi author Bruce Sterling explains that "design fiction doesn't tell stories - instead, it designs prototypes that imply a changed world," therefore "design fiction is a creative act that puts the viewer into a different conceptual space - for a while." Here, we might ask *how different*, as the degree becomes a crucial parameter. Moreover, one of the most prominent aims of such an act is disclosed by Sterling in his informal definition of design fiction itself, as the "deliberate use of diegetic prototypes to suspend disbelief about change" (Sterling, 2013).

The expression "suspension of disbelief," which is often used in this design field, is borrowed from literature, where in the English language it was introduced by Coleridge, suggesting that the narrative component is an integral part of the process and of its success. More importantly, it points to the concept of belief, which has to be pursued with specific strategies of persuasion and manipulation of the audience's perception.

Dunne & Raby, who are central figures in the entire development of speculative design, suggest concentrating on the outer boundaries of plausibility as one of the most fertile parts of such spectrum, where the fringe condition when it is not easy to identify what might be real and what not, might produce a *productive confusion*.

7 VIRTUAL SPACES AS DIEGETIC PROTOTYPES

It's not by chance that Sterling's definition recalls another key concept, the "diegetic prototype," borrowed from cinema, where it is an object that does not exist in actual reality, but is fully functioning within the space of the story (the *diegesis*). In design fiction, the story, the "changed world," is an intentional deviation from reality, whether for critique, provocation, or else.

Science Fiction is the most typical environment for diegetic prototypes and a very clear example is the "gesture-based computer interface technology" featured in the movie "Minority Report" (2001), developed by computer engineer John Underkoffler. Unlike other futuristic technologies featured in cinema this was fully developed, with a specific "dictionary of gestures" and dedicated "training videos and manuals", giving it "technological consistency" to achieve "technological sincerity," becoming "part of the film's natural landscape" (Kirby, 2010).

If technological objects are clear conventional examples of diegetic prototypes, we could wonder if architectural space itself could act as such.

A clear example is provided by spatial installation art, where 1:1 spaces are crafted to *literally* immerse the audience inside the diegesis. Various works of the artist duo Jonah Freeman + Justin Lowe could represent quite well this category, as in the case of Floating Chain (2014), which media theorist Shannon Mattern has described as "part fun-house, part anthropological diorama", underlining both ethnographic and archaeological qualities to it (Mattern, 2014).

This form of spatial narrative explorations can be extremely eloquent but requires necessarily a considerable amount of resources to be realized, hence it is a useful comparative example, but cannot become a widespread tool. Its performance as a diegetic prototype also depends on whether it can perform believably.

That's why an immediate question arises: could immaterial versions of full-scale narrative dioramas, as constructed virtual spaces, fit the purpose of acting as diegetic prototypes?

The crucial point is whether it could be utilized with similar aims, notably to "contextualize [the prototype] in the social sphere", giving it also a political dimension (Kirby, 2010).

Mattern suggests that it's the approach and the qualities that these installation spaces have that could help in that sense, as these dioramas become means to investigate the "underlying

rules and hidden structures — the epistemes — that delimit the realm of thought", "speculat-[ing] as to what those hidden structures might be" (Mattern, 2014).



Figures 2-5. Photos and rendering recontextualization of "Hardware Stories" – Animali Domestici with Jakob Sieder-Semlitsch + Lynn Hyun Kieffer – 2022. Source: Prototypes and images by the authors – Hi-res link QR code or: https://imgur.com/a/2HrZHes.

8 TWISTING THE EVERYDAY: HARDWARE STORIES

The first part of the project "Hardware Stories" is constituted by the 1:1 material prototype of a floor made of various highly heterogeneous tiles (Figure 5) and a series of custom-made tools to craft and maintain them (Figure 4), developed for the exhibition "Architectures of cohabitation" in Berlin in May 2022, curated by ARCH+. Its underlying speculative narrative, imagined within the context of Scandinavian suburbia, where free time availability is a cherished collective political achievement, suggests that the creation of particular hosting environments for other species could become a sort of hobby. Hinting at the diffused DIY culture as a form of social bond, it is rooted in very mundane everyday situations, that were very visible to the authors.

For example, some tiles are made with sponges filled with paper porcelain, creating a fragile ceramic maze, and others are cast with watering jelly beads, that shrink over time, and present a network of spherical cavities that were later filled with cake decorative sugars to attract certain types of spiders (Figure 3). Travertine leftovers and offcuts were scratched and sand-blasted to generate a micro-porosity on the outer surface for snails to collect calcium.

But to proceed to a more subtle level of storytelling and envision the interactions that the prototype would generate, in a successive moment, the tiles have been later recontextualized in a very familiar space, highly inspired by the authors' household, to help imagine them within specific daily routines, habits, and movements (Figure 2). This moment of very informal self-ethnography allowed us to start outlining a layer of anticipatory storytelling. Knowing by experience the spaces and moments of friction that an environment like the kitchen produces when preparing a meal, allows us to reimagine each component of the project performing differently.

9 ANTICIPATORY ETHNOGRAPHY

A key aspect in the operational use of storytelling is that it can support both the phase of initial conception of a design experiment and the later moment of critical analysis of it, where it is key to focus on small details and nuances, key in fragile equilibriums, like multispecies coexistence.

A possible strategy to achieve that focus is to pursue a situated experience, but in the absence of a reality that can be directly perceived, the "diegesis [can] serve to 'situate via

proxy'." This is one of the core modalities of "anticipatory ethnography", meaning "to immerse oneself in a design fiction with a design ethnography 'hat' on and gathering field notes" (Lindley et al, 2014).

Obviously, it is important to start with a disclaimer, clarifying that it is a different thing from conventional ethnography, which involves the direct interaction with the people or beings that are studied, through so-called participant observation. The "field notes" gathered from an immersion in the design fiction constitute, effectively, just a particular process of storytelling, which utilizes techniques inspired by the extensive toolkit developed by ethnographers. Whether that might be media analysis or a form of literary criticism, it doesn't seem relevant here to delve into terminology, but it's useful to note that it attempts to take an internal perspective and it is part of the design process, rather than being an external evaluation of it.

Within a more-than-human dimension, such internal perspective is not only the humans', but should include the "voices" of "multiple species of plants, animals, fungi, and microbes", as attempted by "multispecies ethnography", which looks at how "how diverse organisms are entangled in political, economic, and cultural systems" (Gatto & McCardle, 2019).

In this sense, it's a highly elaborate definition of both the story and its representations that allow us to look much closer at many details, breaking down interactions into very small steps and imagining also some forms of micro-interactions that might appear invisible at first, as with microbiota, and where the frictions are less obvious.

In order to scaffold and properly structure the process of imagination, storytelling becomes a form of simulation: creating a reality, developing it in detail, and thinking through its internal dynamics (both in words and visual media) as if they were normalized.

Normalization plays, in fact, a key role in multiple elements of design fiction, as noted by Kirby regarding diegetic prototypes, which "are at once both completely artificial … and normalized … as practical objects that … people actually use as everyday objects" (Kirby, 2010).



Figures 6-8. Renderings of "Beyond Stop Building - Alive" – Animali Domestici - 2024. Source: Images by the authors – Hi-res link QR code or: https://imgur.com/a/o53IqVu.

10 NEW TECHNO-SOCIAL PARADIGM: BEYOND STOP BUILDING - ALIVE

As part of a contribution to a journal issue on the topic of "Stop Building", the authors proposed to raise a critical conversation by showcasing three housing projects within speculative scenarios that would pursue an extreme reduction of environmental impact in slightly different ways.

The scenarios were carefully constructed based on small deviations from the recent Danish regulation on Life Cycle Assessment (LCA). The example included here, identified by the motto

"Building Alive", suggested that LCA parameters would include biodiversity conservation and support in the equation, to overcome the tendency to carbon tunnel vision (Figure 8).

The way residents' lives would relate to those spaces was, on the other hand, narrated through a very informal storytelling mechanism: through newspaper articles and casual images of ordinary situations. Articles about specific architecture projects were transposed to the new speculative spaces, through a process of rewriting, inheriting impressions and nuances nested within the dialogues and anecdotal personal accounts.

In this way, a much more situated description started to emerge, where additional fictional anecdotes, relating to specific interactions with nonhumans could be easily and organically inserted. Without needing to visualize the moment of friction, some images hint at indexical relationships that suggest them, attempting to be as subtle as possible in that visual narrative.

In the basin shot, for instance, a moment of bathroom routine before breakfast is perhaps suggested by the presence of a drink and a pastry on the low table (Figure 6), which is likely to appeal to the multiple crawling insects that inhabit the micro-landscapes of the walls, which contain a layer of soil that nurtures plants and mosses, hosts multiple crawling insects (Figure 7). Here a popular DIY solution mediates that encounter, as the table legs are sitting in small cups filled with water and oil, preventing any crawling creature from accessing the table.

11 CONCLUSIONS

To identify alternative forms of multispecies coexistence in which the boundaries between the human and nonhuman spheres are clearly blurred and constantly co-produced, architectural experimentation should focus on spaces of mutual interaction between different species. As a result, the nuances of such relationships, as well as the frictions they generate, are critical components of these types of investigation.

Because direct experience may be difficult to collect and study, design fiction may provide a valid alternative entry point into the field, with the construction of detailed virtual spaces and their speculative ethnographic accounts allowing for the simulation of situated experiences.

It is thus critical to identify and develop design tools that allow us to capture and act on subtle nuances in all aspects and stages of the design process, such as observation, imagination, storytelling, and visualisation.

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